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There is a large variety of battery chargers in the market. But linear chargers with mains power transformers are heavy, expensive and dissipate lot of heat. Chargers with pulsed technologies are relatively expensive and create a large spectrum of electromagnetic noise.

Charging current is selected with switches S1 through S4. It can be controlled before the bridge rectifier comprising diodes D1 through D4, or after the bridge. For that purpose, voltmeters should be connected to connectors CON2 and/or CON3. Battery under charge (BUC) is connected between CON5 and CON6 using a two-wire cable with a crocodile clip.

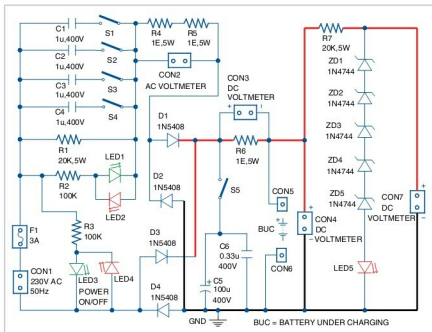


Fig. 1: Circuit diagram of the simple low-cost and versatile battery charger

LED3 and LED4 are used as power on/off indicators.

- Crocodile clip for battery terminal connections
- Voltmeter

An actual-size, single-side PCB layout for the simple low-cost, versatile battery charger is shown in Fig. 2